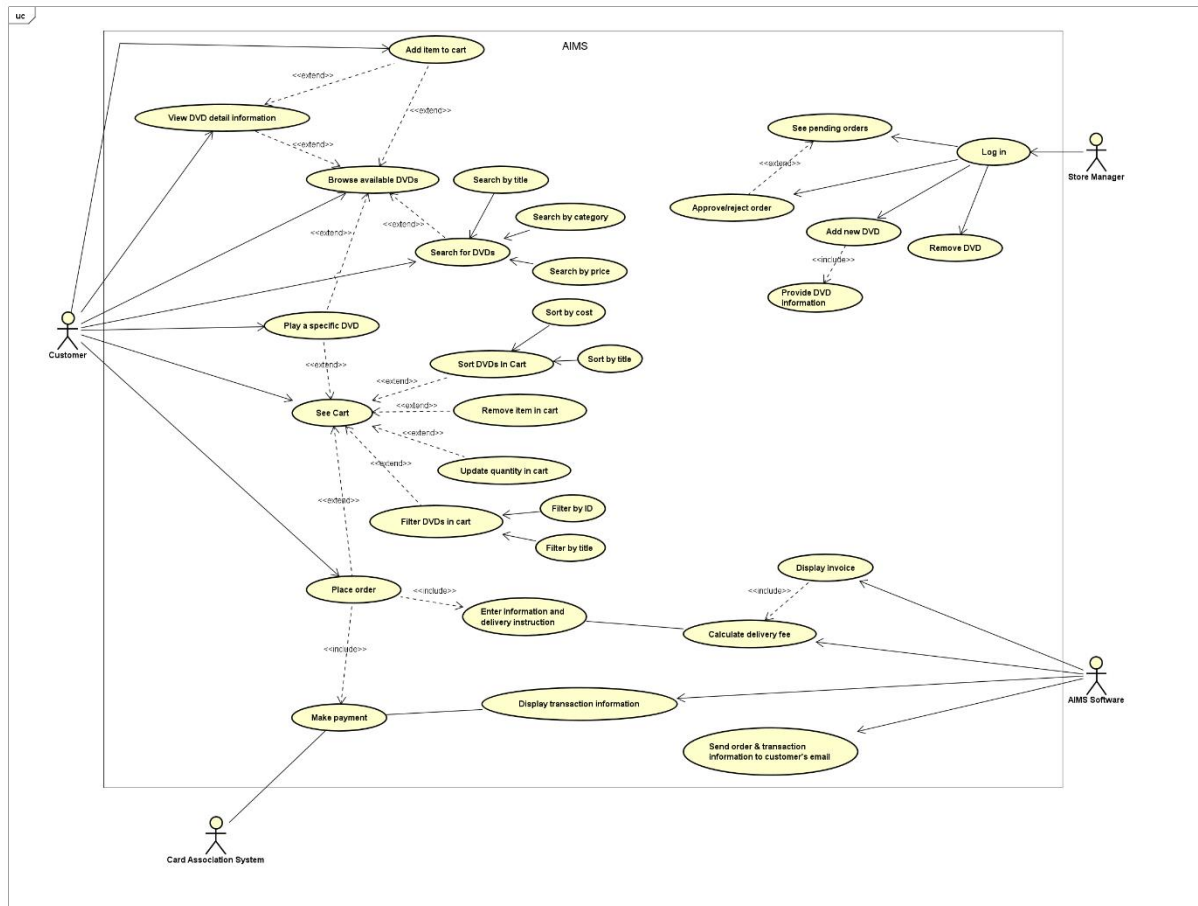


Report Lab05

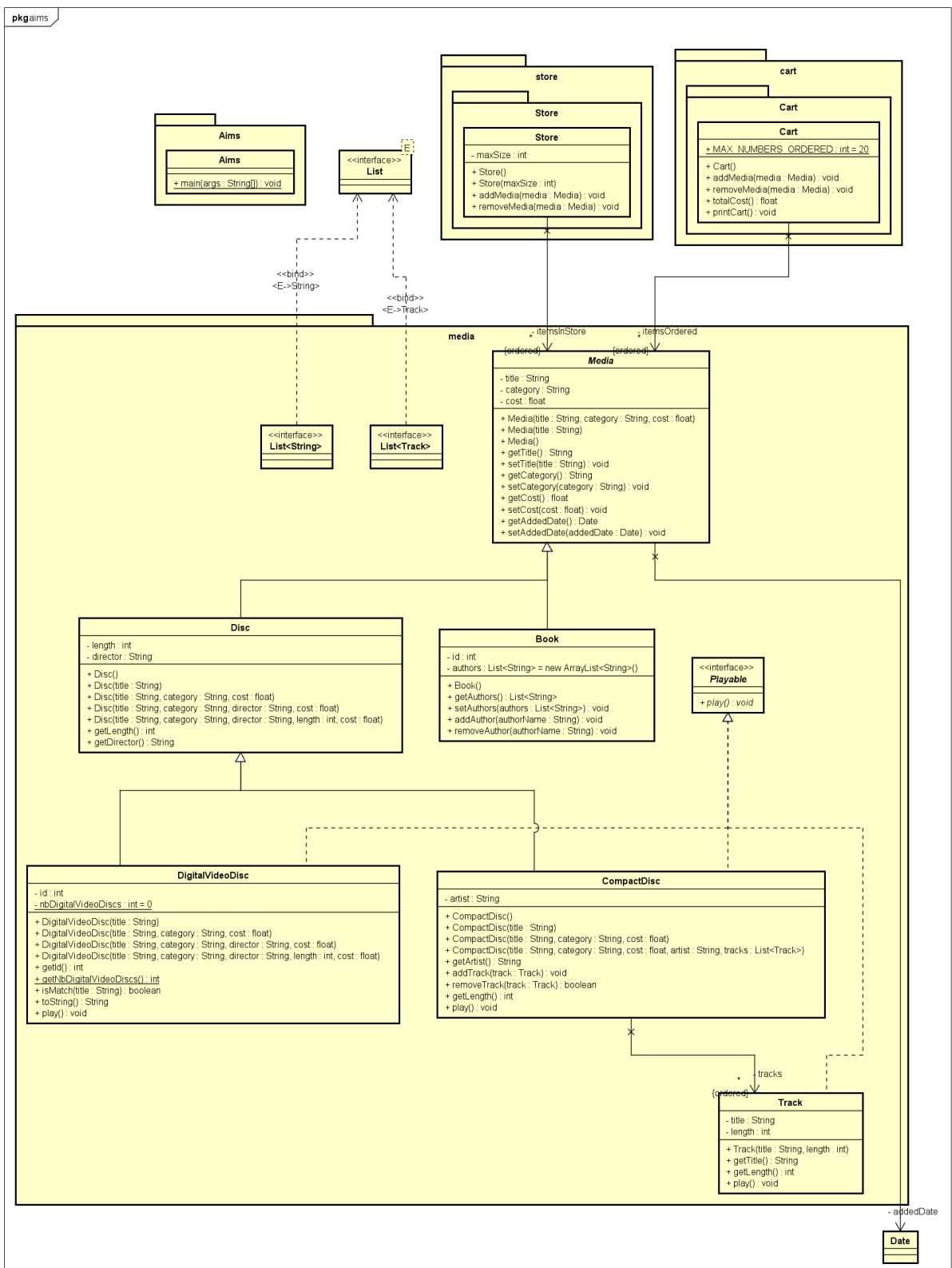
Name: Ngô Văn Đông

Student ID: 20235486

Use-case diagram



Class Diagram



1. Swing components

1.1. AWTAccumulator

```
public class AWTAccumulator extends Frame {
    private TextField tfInput;
    private TextField tfOutput;
    private int sum = 0;

    // Constructor to setup the GUI components and event handlers
    public AWTAccumulator() {
        setLayout(new GridLayout(2,2));
        add(new Label("Enter an integer: "));
        tfInput = new TextField(10);
        add(tfInput);
        tfInput.addActionListener(new TFInputListener());

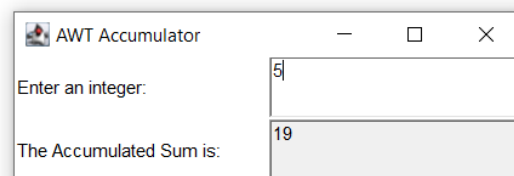
        add(new Label("The Accumulated Sum is: "));

        tfOutput = new TextField(10);
        tfOutput.setEditable(false);
        add(tfOutput);

        setTitle("AWT Accumulator");
        setSize(350,120);
        setVisible(true);
    }

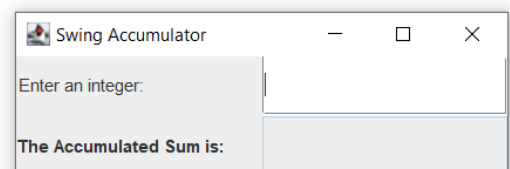
    public static void main(String [] args) {
        new AWTAccumulator();
    }

    private class TFInputListener implements ActionListener {
        @Override
        public void actionPerformed(ActionEvent evt) {
```



1.2. SwingAccumulator

```
1 package hust.soict.dsai.swing;
2
3 import java.awt.Container;
4
5 public class SwingAccumulator extends JFrame {
6     private JTextField tfInput;
7     private JTextField tfOutput;
8     private int sum = 0;
9
10    // Constructor to setup the GUI components and event handlers
11    public SwingAccumulator() {
12        Container cp = getContentPane();
13        cp.setLayout(new GridLayout(2,2));
14        cp.add(new Label("Enter an integer: "));
15        tfInput = new JTextField(10);
16        cp.add(tfInput);
17        tfInput.addActionListener(new TFInputListener());
18
19        cp.add(new JLabel("The Accumulated Sum is: "));
20
21        tfOutput = new JTextField(10);
22        tfOutput.setEditable(false);
23        cp.add(tfOutput);
24
25        setTitle("Swing Accumulator");
26        setSize(350,120);
27        setVisible(true);
28    }
29 }
```



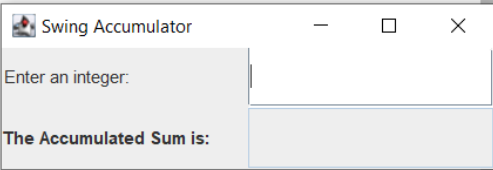
```

    }

    public static void main(String [] args) {
        new SwingAccumulator();
    }

    private class TFInputListener implements ActionListener {
        @Override
        public void actionPerformed(ActionEvent evt) {
            int numberIn = Integer.parseInt(tfInput.getText());
            sum += numberIn;
            tfInput.setText("");
            tfOutput.setText(sum + "");
        }
    }
}

```



2. Organizing Swing components with Layout Managers

```

package hust.soict.dsai.swing;

import java.awt.BorderLayout;

public class NumberGrid extends JFrame{
    private JButton[] btnNumbers = new JButton[10];
    private JButton btnDelete, btnReset;
    private JTextField tfDisplay;

    public NumberGrid() {

        tfDisplay = new JTextField();
        tfDisplay.setComponentOrientation(ComponentOrientation.RIGHT_TO_LEFT);

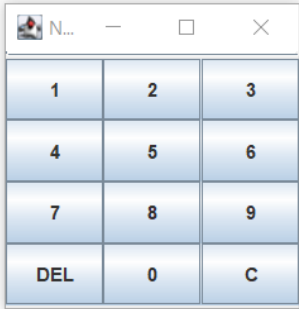
        JPanel panelButtons = new JPanel(new GridLayout(4, 3));
        addButtons(panelButtons);

        Container cp = getContentPane();
        cp.setLayout(new BorderLayout());
        cp.add(tfDisplay, BorderLayout.NORTH);
        cp.add(panelButtons, BorderLayout.CENTER);

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setTitle("Number Grid");
        setSize(200, 200);
        setVisible(true);
    }

    private class ButtonListener implements ActionListener{
        @Override
        public void actionPerformed(ActionEvent e) {
            String button = e.getActionCommand();

```



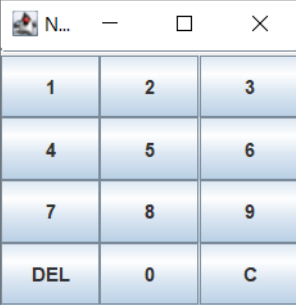
```

private class ButtonListener implements ActionListener{
    @Override
    public void actionPerformed(ActionEvent e) {
        String button = e.getActionCommand();
        if (button.charAt(0) >= '0' && button.charAt(0) <= '9') {
            tfDisplay.setText(tfDisplay.getText() + button);
        } else if (button.equals("DEL")) {
            if (tfDisplay.getText().length() >= 1) {
                tfDisplay.setText(tfDisplay.getText().substring(0, tfDisplay.getText().length()-1));
            }
        } else if (button.equals("C")) {
            tfDisplay.setText("");
        } else {
            tfDisplay.setText(tfDisplay.getText());
        }
    }
}

void addButtons(JPanel panelButtons) {
    ButtonListener btnListener = new ButtonListener();
    for (int i = 1; i<=9; i++) {
        btnNumbers[i] = new JButton(""+i);
        panelButtons.add(btnNumbers[i]);
        btnNumbers[i].addActionListener(btnListener);
    }
    btnDelete = new JButton("DEL");
    panelButtons.add(btnDelete);
    btnDelete.addActionListener(btnListener);

    btnNumbers[0] = new JButton("0");
    panelButtons.add(btnNumbers[0]);

```



1	2	3
4	5	6
7	8	9
DEL	0	C

```

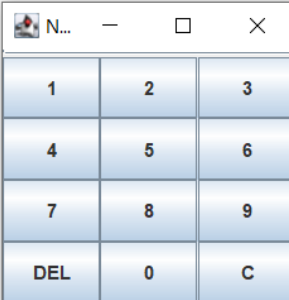
void addButtons(JPanel panelButtons) {
    ButtonListener btnListener = new ButtonListener();
    for (int i = 1; i<=9; i++) {
        btnNumbers[i] = new JButton(""+i);
        panelButtons.add(btnNumbers[i]);
        btnNumbers[i].addActionListener(btnListener);
    }
    btnDelete = new JButton("DEL");
    panelButtons.add(btnDelete);
    btnDelete.addActionListener(btnListener);

    btnNumbers[0] = new JButton("0");
    panelButtons.add(btnNumbers[0]);
    btnNumbers[0].addActionListener(btnListener);

    btnReset = new JButton("C");
    panelButtons.add(btnReset);
    btnReset.addActionListener(btnListener);
}

public static void main(String[] args) {
    new NumberGrid();
}

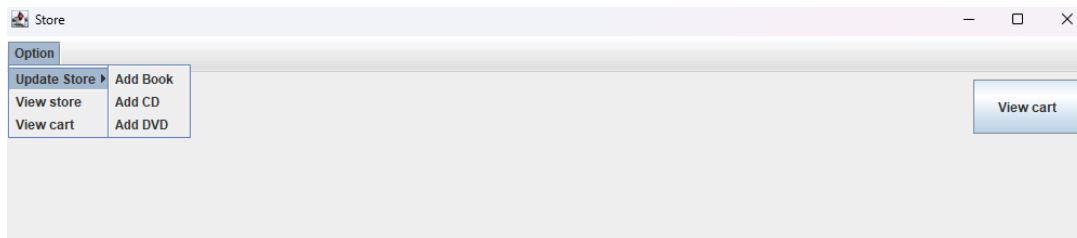
```



1	2	3
4	5	6
7	8	9
DEL	0	C

3. Create a graphical user interface for AIMS with Swing

3.1. View Store Screen



3.2. Adding more user interaction

```

JButton addCartBtn = new JButton("Add to Cart");
addCartBtn.addActionListener(new ActionListener() {

    @Override
    public void actionPerformed(ActionEvent e) {
        try {
            cart.addMedia(media);
        } catch (LimitExceededException e1) {
            // TODO Auto-generated catch block
            e1.printStackTrace();
        }
    }
});
container.add(addCartBtn);

if (media instanceof Playable) {
    JButton playBtn = new JButton("Play");
    playBtn.addActionListener(new ActionListener() {
        @Override
        public void actionPerformed(ActionEvent e) {
            JDialog playDialog = new JDialog();
            JPanel mainGui = new JPanel(new BorderLayout());
            mainGui.setBorder(new EmptyBorder(20, 20, 20, 20));

            // Display Playing Message
            mainGui.add(new JLabel("Playing... " + media.getTitle()), BorderLayout.CENTER);
            System.out.println(media.getTitle());
            // Close Button
            JPanel buttonPanel = new JPanel(new FlowLayout());
            JButton close = new JButton("Stop");
            close.addActionListener(ev -> {
                playDialog.setVisible(false);
                System.out.println("Stopped playing.");
            });
            buttonPanel.add(close);
            mainGui.add(buttonPanel, BorderLayout.SOUTH);

            mainGui.add(buttonPanel, BorderLayout.SOUTH);

            playDialog.setContentPane(mainGui);
            playDialog.setLocationRelativeTo(playBtn);
            playDialog.pack();

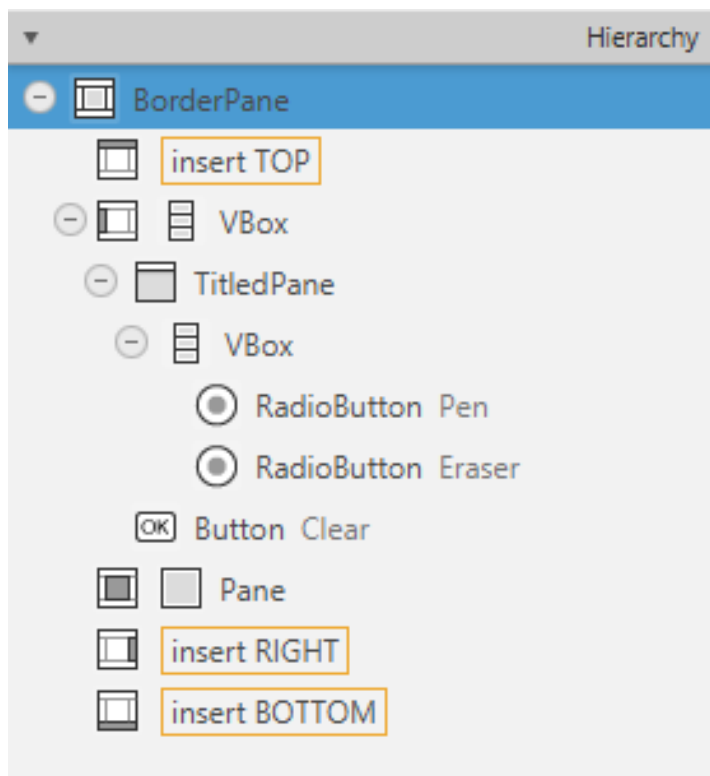
            // Show Dialog
            playDialog.setVisible(true);
        }
    });
    container.add(playBtn);
}

this.add(Box.createVerticalGlue());
this.add(title);
this.add(cost);
this.add(Box.createVerticalGlue());
this.add(container);

this.setBorder(BorderFactory.createLineBorder(Color.BLACK));
}

```

4. JavaFX API



Inspector

Q

⚙

▶

Properties : BorderPane

▼

Layout : BorderPane

Internal

Padding

☐

8

>

☐

8

☐

8

☐

8

Size

Min Width

USE_PREF_SIZE

▼

Min Height

USE_PREF_SIZE

▼

Pref Width

640

▼

Pref Height

480

▼

Max Width

USE_PREF_SIZE

▼

Max Height

USE_PREF_SIZE

▼

Width

640

Height

480

Position

Layout X

0

Layout Y

0

Transforms

Rotate

0

Rotation Axis

x

0

y

0

z

1

Scale X

1

Scale Y

1

Scale Z

1

Translate X

0

Translate Y

0

Translate Z

0

Bounds

Layout Bounds

0,0 640x480

▼

Bounds In Local

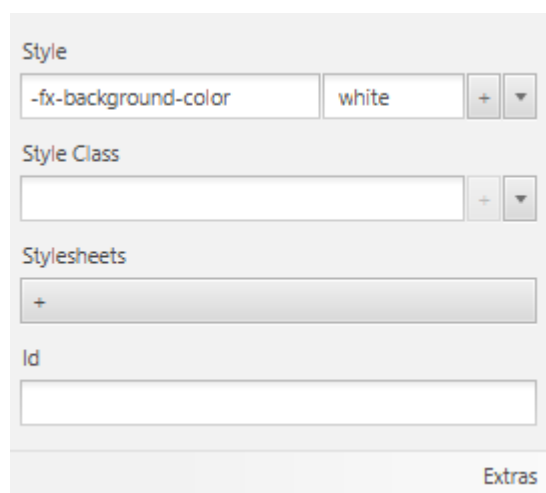
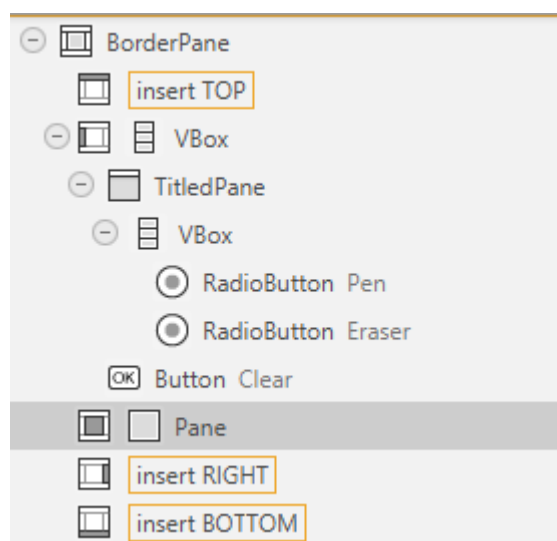
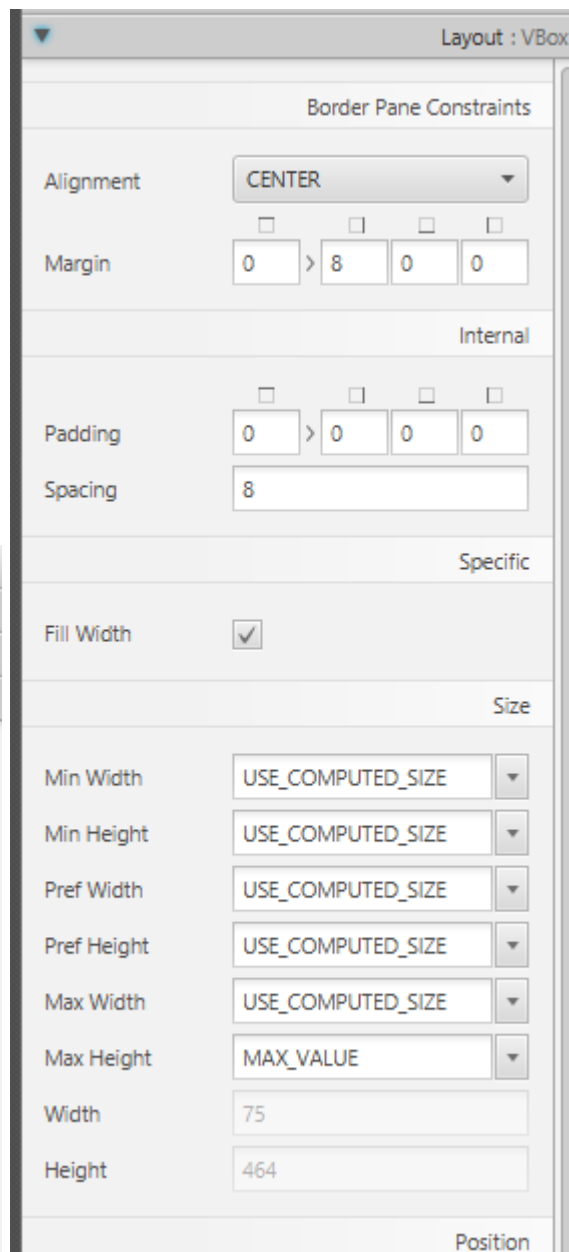
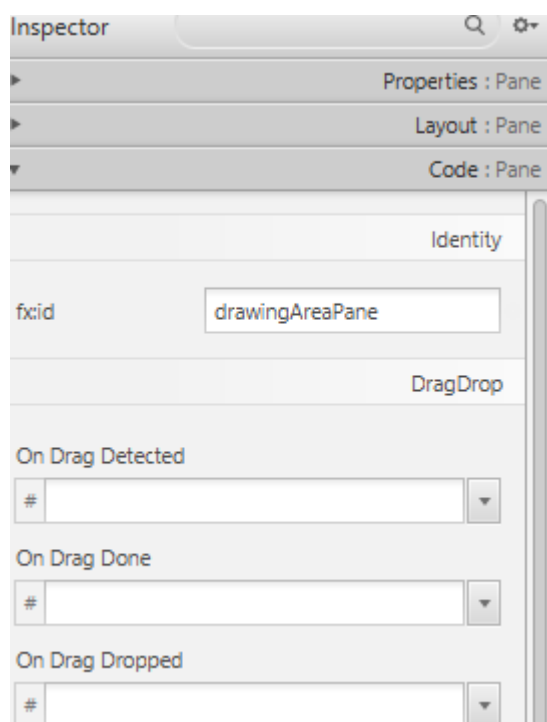
0,0 640x480

▼

Bounds In Parent

0,0 640x480

▼



On Context Menu Requested

#

On Mouse Clicked

#

On Mouse Dragged

#

On Mouse Entered

#

On Mouse Exited

BorderPane

insert TOP

VBox

TitledPane

VBox

RadioButton Pen

RadioButton Eraser

OK Button Clear

Pane

insert RIGHT

insert BOTTOM

Properties : Button

Text

Text

Font

Text Fill

Wrap Text ☐

Text Alignment

Text Overrun

Ellipsis String

Underline ☐

Line Spacing

Layout : Button

VBox Constraints

Vgrow

Margin ☐ 0 > ☐ 0 ☐ 0 ☐ 0

Internal

Padding ☐ 0 > ☐ 0 ☐ 0 ☐ 0

Size

Min Width

Min Height

Pref Width

Pref Height

Max Width

Max Height

Width

Height

Code : Button

On Action
#

DragDrop

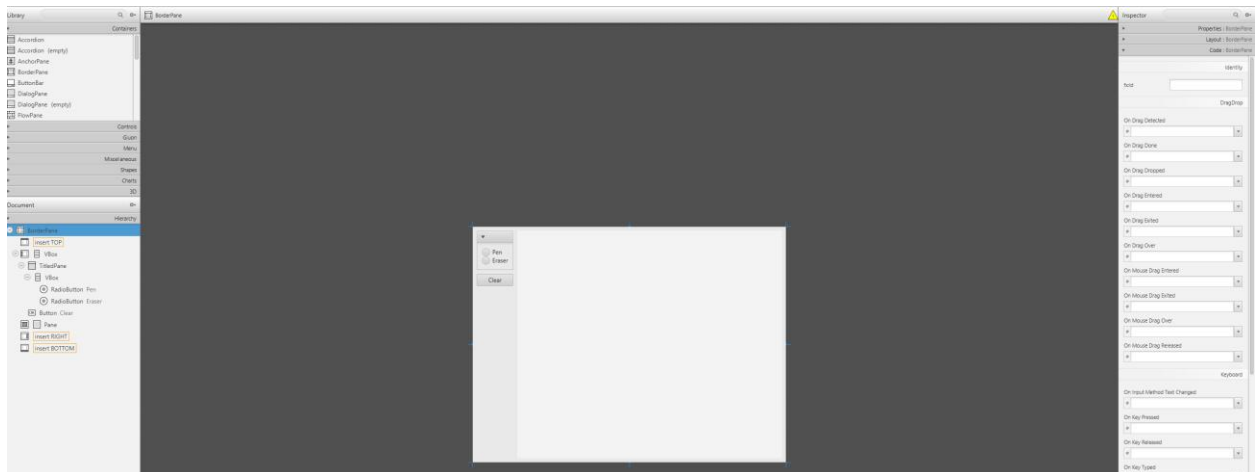
On Drag Detected
#

On Drag Done
#

On Drag Dropped
#

On Drag Entered
#

On Drag Exited



```

//img= mouseClicked.java,
import javafx.event.ActionEvent;

public class PainterController {

```

```

    private int color;

    @FXML
    private ToggleGroup identity;

    @FXML
    void chooseOption(ActionEvent event) {
        String button = ((RadioButton)event.getSource()).getText();
        if (button.equals("Pen")) {
            System.out.println("1");
            color = 1;
        }else {}
        System.out.println("0");
        color = 0;
    }

    @FXML
    private Pane drawingAreaPane;

    @FXML
    void clearButtonPressed(ActionEvent event) {
        drawingAreaPane.getChildren().clear();
    }

```

```

import javafx.scene.control.ToggleGroup;
<?import javafx.scene.layout.BorderPane?>
<?import javafx.scene.layout.Pane?>
<?import javafx.scene.layout.VBox?>

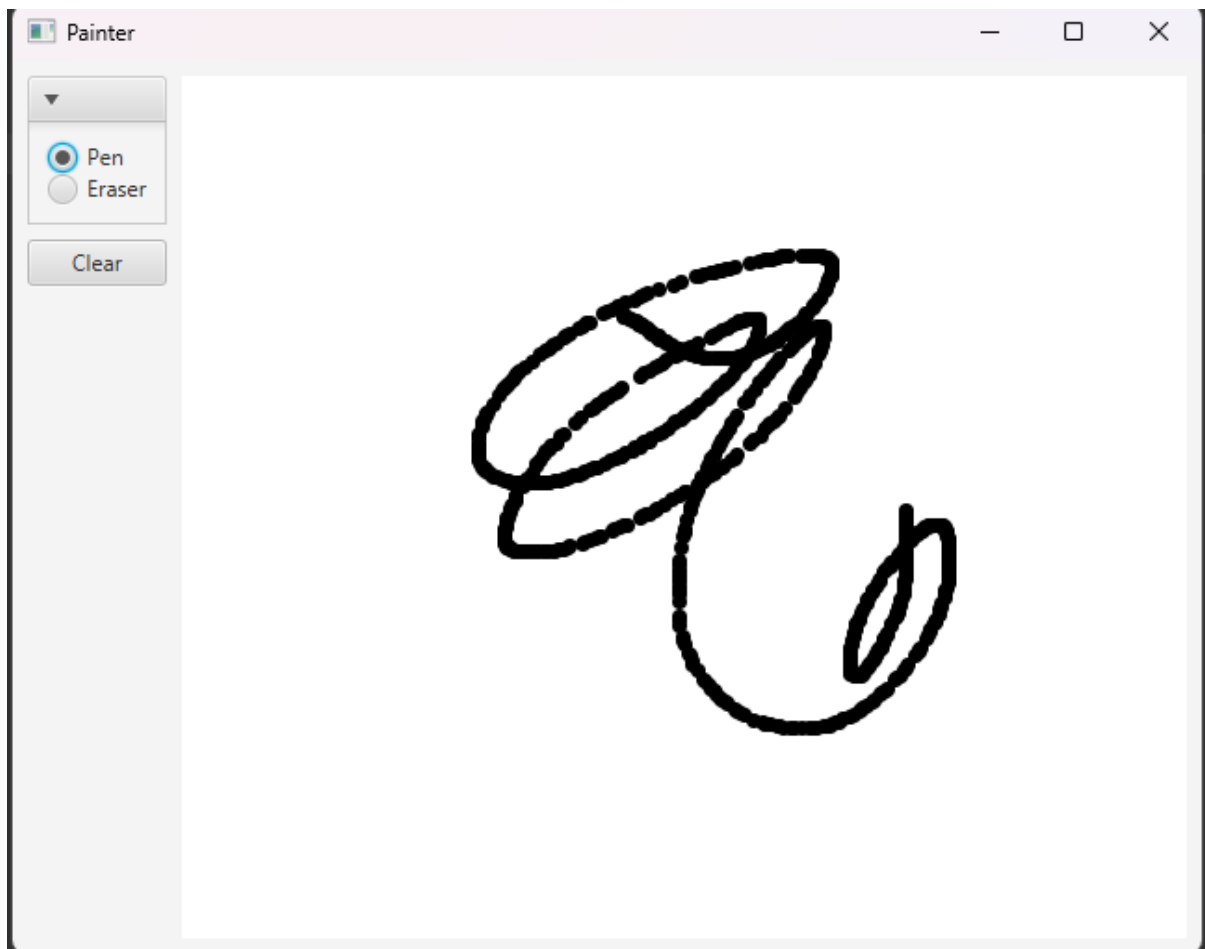
```

Bind to grammar/schema...

```

<BorderPane maxHeight="-Infinity" maxWidth="-Infinity" minHeight="-Infinity" minWidth="-Infinity" prefHeight="480.0" prefWidth="640.0">
    <left>
        <VBox maxHeight="1.7976931348623157E308" spacing="8.0" BorderPane.alignment="CENTER">
            <BorderPane.margin>
                <Insets right="8.0" />
            </BorderPane.margin>
            <children>
                <TitledPane>
                    <content>
                        <VBox>
                            <children>
                                <RadioButton mnemonicParsing="false" onAction="#chooseOption" text="Pen">
                                    <toggleGroup>
                                        <ToggleGroup fx:id="identity" />
                                    </toggleGroup>
                                </RadioButton>
                                <RadioButton mnemonicParsing="false" onAction="#chooseOption" text="Eraser" toggleGroup="$identity" />
                            </children>
                        </VBox>
                    </content>
                </TitledPane>
                <Button maxWidth="1.7976931348623157E308" mnemonicParsing="false" onAction="#clearButtonPressed" text="Clear" />
            </children>
        </VBox>
    </left>
    <center>
        <Pane fx:id="drawingAreaPane" onMouseDragged="#drawingAreaMouseDragged" style="-fx-background-color: white;" BorderPane.alignment="center">
            <padding>
                <Insets bottom="8.0" left="8.0" right="8.0" top="8.0" />
            </padding>
        </Pane>
    </center>
</BorderPane>

```



Handle Erase Function

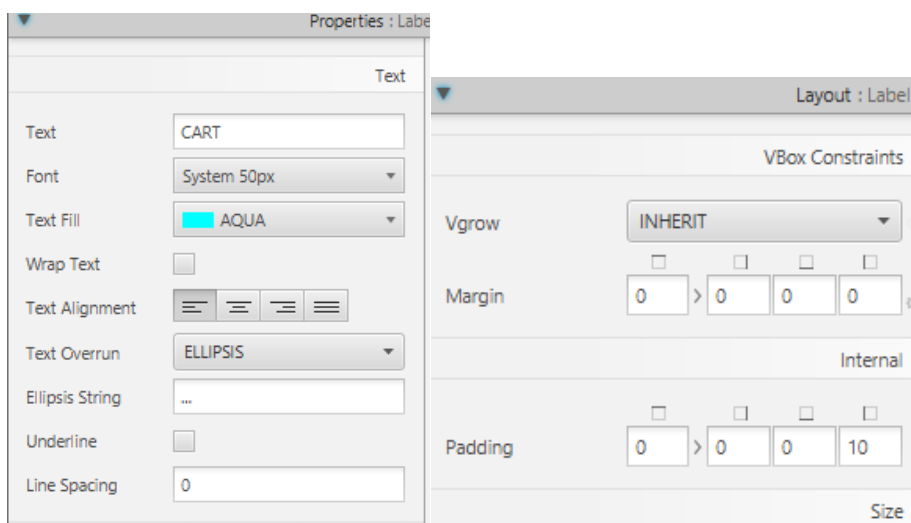
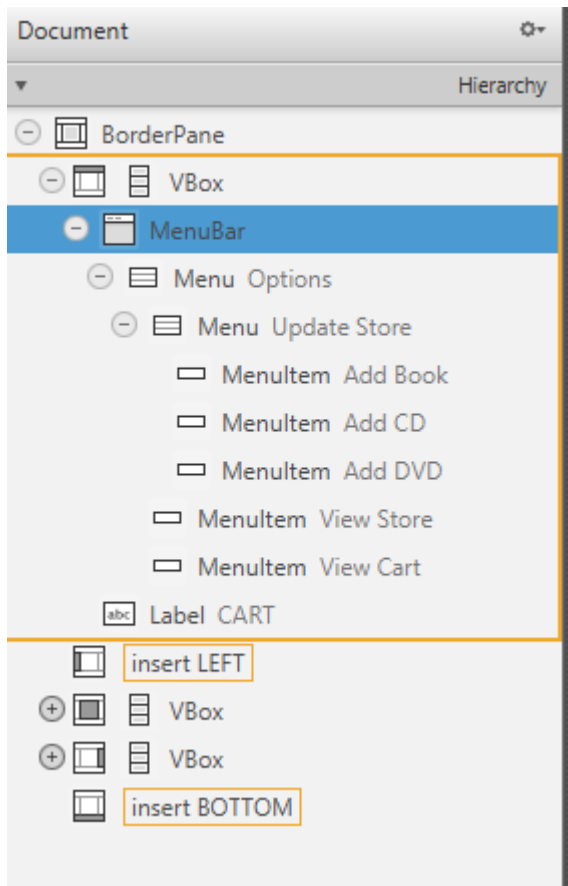
```

@FXML
void drawingAreaMouseDragged(MouseEvent event) {
    if (color == 1 && event.getX() >= 0) {
        Circle newCircle = new Circle(event.getX(), event.getY(), 4.0, Color.BLACK);
        drawingAreaPane.getChildren().add(newCircle);
    } else if (color == 0 && event.getX() >= 0) {
        Circle newCircle = new Circle(event.getX(), event.getY(), 4.0, Color.WHITE);
        drawingAreaPane.getChildren().add(newCircle);
    }
}

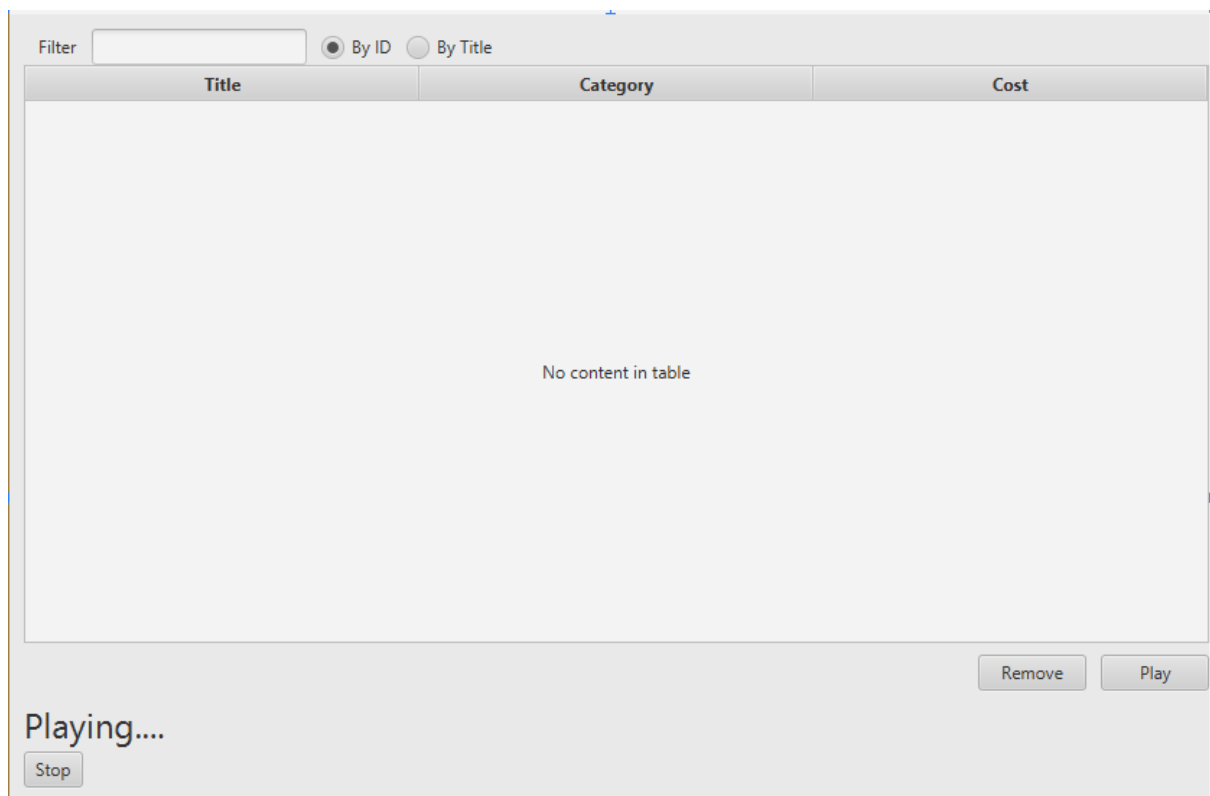
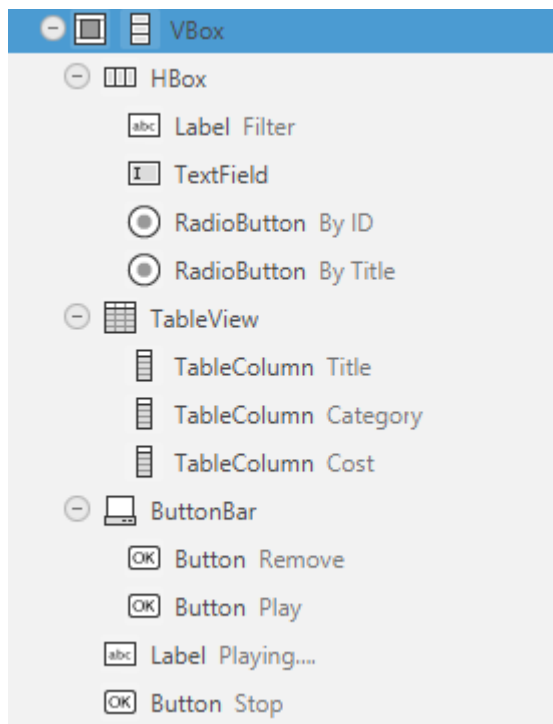
```

5. Setting up the View Cart Screen with ScreenBuilder

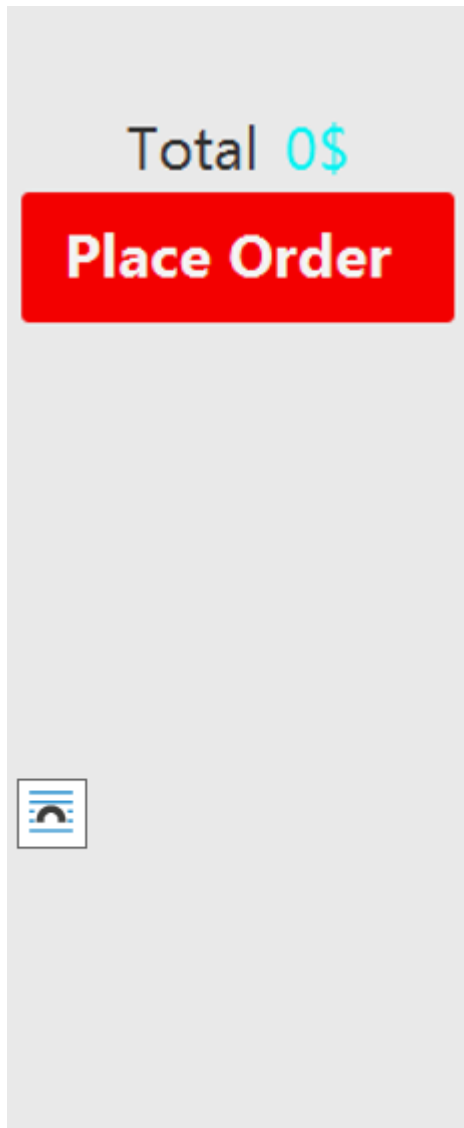
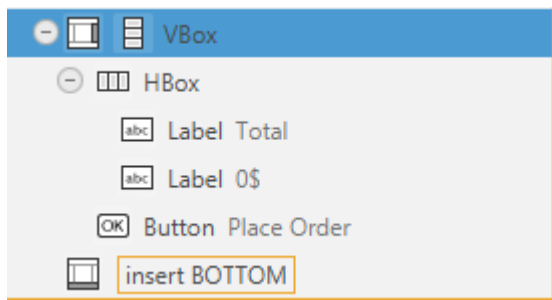
5.2. Setting up the TOP area



5.3. Setting up the CENTER area



5.4. Setting up the RIGHT area



6. Integrating JavaFX into Swing application – the JFXPanel class

```

package hust.soict.dsai.aims.screen;

import java.awt.Dimension;

public class CartScreen extends JFrame {

    private Cart cart;
    private ControllerScreen controllerScreen;

    public CartScreen(Cart cart, ControllerScreen c) {
        super();

        this.cart = cart;

        JFXPanel fxPanel = new JFXPanel();
        this.add(fxPanel);
        this.setTitle("Cart");
        // this.setVisible(true);
        this.setPreferredSize(new Dimension(1024, 768));
        pack();
        Platform.runLater(new Runnable() {

            @Override
            public void run() {
                try {
                    FXMLLoader loader = new FXMLLoader(getClass()
                        .getResource("/hust/soict/dsai/aims/screen/cart.fxml"));
                    CartScreenController controller = new CartScreenController(cart, c);
                    loader.setController(controller);
                    Parent root = loader.load();
                    fxPanel.setScene(new Scene(root, 1024, 768));
                } catch (IOException e) {
                    e.printStackTrace();
                }
            }
        });
    }
}

```

8. Deleting a media

```

@FXML
void btnRemovePressed(ActionEvent event) {
    Media media = tblMedia.getSelectionModel().getSelectedItem();
    cart.removeMedia(media);
    totalCost.setText(cart.totalCost()+"$");
}

```

9. Filter items in cart – FilteredList

```

void showFilterMedia(String searchString) {
    if(searchString.isEmpty()) {
        tblMedia.setItems(this.cart.getItemsOrdered());
    } else {
        if(radioBtnFilterId.isSelected()) {
            tblMedia.setItems(new FilteredList<Media>(this.cart.getItemsOrdered(), item-> item.getId()==Integer.parseInt(searchString)));
        } else {
            tblMedia.setItems(new FilteredList<Media>(this.cart.getItemsOrdered(), item-> item.getTitle().contains(searchString)));
        }
    }
}

```

10. Complete the Aims GUI application

AddItemToStoreScreen

```

package hust.soict.dsai.aims.screen;

import java.awt.BorderLayout;

public class AddItemToStoreScreen extends JFrame {
    protected Store store;
    protected Cart cart;
    protected JTextField title;
    protected JTextField category;
    protected JTextField cost;
    protected JButton addBtn;
    protected ControllerScreen c;
    protected JPanel center;
    protected int numberInput = 3;

    JMenuBar createMenuBar() {
        JMenu menu = new JMenu("Option");

        JMenu smUpdateStore = new JMenu("Update Store");
        JMenuItem addBookMenu = new JMenuItem("Add Book");
        addBookMenu.addActionListener(e -> {
            c.showAddBookScreen();
        });
        smUpdateStore.add(addBookMenu);
        JMenuItem addCDMenu = new JMenuItem("Add CD");
        addCDMenu.addActionListener(e -> {
            c.showAddCDCreen();
        });
        smUpdateStore.add(addCDMenu);
        JMenuItem addDVDMenu = new JMenuItem("Add DVD");
        addDVDMenu.addActionListener(e -> {
            c.showAddDVDScreen();
        });
        smUpdateStore.add(addDVDMenu);
    }
}

```

AddDigitalVideoDiscToStoreScreen

```

package hust.soict.dsai.aims.screen;

import java.awt.Dimension;

public class AddDigitalVideoDiscToStoreScreen extends AddItemToStoreScreen {
    private JTextField director;
    private JTextField length;

    public AddDigitalVideoDiscToStoreScreen(Store store, Cart cart, ControllerScreen c) {
        super(store, cart, c);
    }

    @Override
    void updateAdd(JPanel panel) {
        this.numberInput = 6;
        // SpringLayout layout = new SpringLayout();
        // JPanel p = new JPanel(layout);
        JLabel directorLabel = new JLabel("director", JLabel.TRAILING);
        panel.add(directorLabel);
        director = new JTextField(2);
        director.setPreferredSize(new Dimension(150, 20));
        directorLabel.setLabelFor(director);
        panel.add(director);
        JLabel lengthLabel = new JLabel("length", JLabel.TRAILING);
        panel.add(lengthLabel);
        length = new JTextField(2);
        lengthLabel.setLabelFor(length);
        panel.add(length);
        JButton tes = new JButton("add");
        tes.setVisible(false);
        panel.add(tes);
        panel.setPreferredSize(new Dimension(100, 300));
        addBtn = new JButton("add");
        addBtn.addActionListener(e -> {
            addMedia();
        });
    }
}

```

AddCompactDiscToStoreScreen

```

package hust.soict.dsai.aims.screen;

import java.awt.Dimension;

public class AddCompactDiscToStoreScreen extends AddItemToStoreScreen {
    private JTextField artist;
    private JTextField listTrack;

    public AddCompactDiscToStoreScreen(Store store, Cart cart, ControllerScreen c) {
        super(store, cart, c);
    }

    @Override
    void updateAdd(JPanel panel) {
        this.numberInput = 6;

        JLabel artistLabel = new JLabel("artist", JLabel.TRAILING);
        panel.add(artistLabel);
        artist = new JTextField(2);
        artist.setPreferredSize(new Dimension(150, 20));
        artistLabel.setLabelFor(artist);
        panel.add(artist);
        JLabel listTrackLabel = new JLabel("List track (each track separated by a comma Ex: track-length)",
            JLabel.TRAILING);

        panel.add(listTrackLabel);
        listTrack = new JTextField(2);
        listTrackLabel.setLabelFor(listTrack);
        panel.add(listTrack);
        JButton tes = new JButton("add");
        tes.setVisible(false);
        panel.add(tes);
        panel.setPreferredSize(new Dimension(100, 300));
        addBtn = new JButton("add");
    }
}

```

AddBookToStoreScreen

```

package hust.soict.dsai.aims.screen;

import java.awt.Dimension;

public class AddBookToStoreScreen extends AddItemToStoreScreen {
    private JTextField listAuthor;

    public AddBookToStoreScreen(Store store, Cart cart, ControllerScreen c) {
        super(store, cart, c);
    }

    @Override
    void updateAdd(JPanel panel) {
        this.numberInput = 5;

        JLabel listAuthorLabel = new JLabel("Authors(Names are separated by a comma)", JLabel.TRAILING);
        panel.add(listAuthorLabel);
        listAuthor = new JTextField(2);
        listAuthor.setPreferredSize(new Dimension(150, 20));
        listAuthorLabel.setLabelFor(listAuthor);
        panel.add(listAuthor);

        JButton tes = new JButton("add");
        tes.setVisible(false);
        panel.add(tes);
        panel.setPreferredSize(new Dimension(100, 300));
        addBtn = new JButton("add");
        addBtn.addActionListener(e -> {
            addMediaToStore();
        });
        panel.add(addBtn);
    }

    public void addMediaToStore() {
        String title = this.title.getText();
    }
}

```

12. Check all the previous source codes to catch/handle/delegate runtime exceptions

```
public void addMedia(Media media) throws LimitExceededException {
    if ((itemsOrdered.size()) >= MAX_ORDERED) {
        throw new LimitExceededException("Full");
    }
    else if (itemsOrdered.contains(media)) {
        System.out.println("This is already in your order!");
    }
    else {
        itemsOrdered.add(media);
        System.out.println("Media added!");
    }
}
```

13. Create a class which inherits from Exception

```
package hust.soict.dsai.aims.exception;

public class PlayerException extends RuntimeException {

    public PlayerException() {
        // TODO Auto-generated constructor stub
    }

    public PlayerException(String message) {
        super(message);
        // TODO Auto-generated constructor stub
    }

    public PlayerException(Throwable cause) {
        super(cause);
        // TODO Auto-generated constructor stub
    }

    public PlayerException(String message, Throwable cause) {
        super(message, cause);
        // TODO Auto-generated constructor stub
    }

    public PlayerException(String message, Throwable cause, boolean enableSuppression, boolean writableStackTrace) {
        super(message, cause, enableSuppression, writableStackTrace);
        // TODO Auto-generated constructor stub
    }
}
```

14. Update the Aims class

```

int choice;
do {
    storeMenu();
    choice = scanner.nextInt();
    scanner.nextLine();

    switch (choice) {
        case 1:
            seeMediaDetails(scanner);
            break;
        case 2:
            addMediaToCart(scanner);
            break;
        case 3:
            playMedia(scanner);
            break;
        case 4:
            seeCurrentCart(scanner);
            break;
        case 0:
            break;
        default:
            System.out.println("Invalid choice! Please choose a number between 0-4.");
    }
} while (choice != 0);
}

```

```

... ..

public static void playMedia(Scanner scanner) {
    System.out.print("Enter the title of the media to play: ");
    String title = scanner.nextLine();
    Media media = store.searchByTitle(title);

    if (media != null) {
        if (media instanceof Playable) {
            if (media instanceof DigitalVideoDisc){
                try {
                    ((DigitalVideoDisc) media).play();
                } catch (PlayerException e) {
                    // TODO Auto-generated catch block
                    e.getMessage();
                    e.toString();
                    e.printStackTrace();
                }
            }
            else if (media instanceof CompactDisc){
                try {
                    ((CompactDisc) media).play();
                } catch (PlayerException e) {
                    // TODO Auto-generated catch block
                    e.getMessage();
                    e.toString();
                    e.printStackTrace();
                }
            }
        } else {
            System.out.println("This media cannot be played.");
        }
    } else {
        System.out.println("Media not found in the store.");
    }
}

```